

Answer Key — Biology Grade 11

Sub-Strand 3.2: Growth and Development in Animals • Lesson 4: Hormones as the Control Switches of Moulting and Metamorphosis

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	A	B	C	D	A	B

USE AFTER: PREDICT PHASE — Predict: groups predict what happens to a caterpillar if the part that sends the 'moult' signal is removed (POE sheet)

Q1. A scientist removes the brain neurosecretory cells (the cells that make Brain Hormone) from a young caterpillar. What do you predict will happen to the caterpillar's moulting?

- A. It will stop moulting because the signal that starts the process is missing**
- B. It will turn into a pupa straight away
- C. It will moult more often because the brain no longer slows growth
- D. It will moult as normal because the shell splits by itself

Correct answer: A. Brain Hormone signals the prothoracic glands to release Ecdysone, which triggers moulting. Without the first signal, no Ecdysone is released and moulting stops.

USE AFTER: OBSERVE PHASE — Part A: 3-row hormone table (Brain Hormone, Ecdysone, Juvenile Hormone) using reference texts and data sheets

Q2. Which gland releases Ecdysone after it receives the signal from Brain Hormone?

- A. Prothoracic glands**
- B. Corpora allata
- C. Malpighian tubules
- D. Salivary glands

Correct answer: A. Brain Hormone from the neurosecretory cells stimulates the prothoracic glands to release Ecdysone. The corpora allata make Juvenile Hormone instead.

USE AFTER: OBSERVE PHASE — Part B: Experiment 2, larva with brain removed then injected with Ecdysone (moult resumes)

Q3. In an experiment, a larva has its brain neurosecretory cells removed and it stops moulting. The same larva is then injected with Ecdysone and it starts to moult again. What does this result show?

- A. Ecdysone is made by the brain neurosecretory cells
- B. Ecdysone triggers moulting and acts after the brain signal**
- C. Moulting does not need any hormones
- D. Ecdysone keeps the larva in the larval form

Correct answer: B. Moulting stopped when the brain signal was removed and resumed when Ecdysone was replaced. This shows Ecdysone is the hormone that triggers moulting and works after the brain signal.

USE AFTER: OBSERVE PHASE — Part B: Experiment 4, larva with corpora allata removed early (small, early pupa)

Q4. In an experiment, the corpora allata (which make Juvenile Hormone) are removed from a young larva. It changes early into a small pupa. What does this show about Juvenile Hormone?

- A. Juvenile Hormone is made by the prothoracic glands
- B. Juvenile Hormone is needed to trigger every moult
- C. Juvenile Hormone keeps the larval state, so low Juvenile Hormone allows pupation**
- D. High Juvenile Hormone makes a larva change into a pupa early

Correct answer: C. With the corpora allata removed, Juvenile Hormone is low. The larva pupates early and stays small. So Juvenile Hormone normally keeps the insect in the larval state, and low levels allow the change to a pupa.

USE AFTER: EXPLAIN PHASE — Explain: CER writing and oral probe 'What would happen if JH stayed high at the last larval moult?'

Q5. A farmer in the Rift Valley has an armyworm larva that is about to make its last larval moult. Suppose Juvenile Hormone stayed high at this moult. What would you expect to happen?

- A. It would moult straight into an adult moth
- B. It would stop moulting because Ecdysone is blocked completely
- C. It would moult into a pupa as normal
- D. It would moult into another larva instead of a pupa**

Correct answer: D. High Juvenile Hormone gives a larva-to-larva moult. Pupation needs low Juvenile Hormone, so the larva would moult into a larger larva instead.

USE AFTER: MODEL BUILDING PHASE — Model: annotating the stepped growth curve from Lesson 3 with the hormone control layer

Q6. A student annotates a stepped growth curve for a grasshopper. Each sudden upward step in body size marks a moult. What should the student write about Ecdysone at each step?

- A. Ecdysone peaks at the step, because it triggers moulting**
- B. Ecdysone is lowest at the step, because it keeps the old cuticle in place
- C. Ecdysone is only made after the insect becomes an adult
- D. Ecdysone is absent during the whole life cycle

Correct answer: A. Ecdysone triggers moulting. The upward steps in the stepped growth curve match the moults, when Ecdysone peaks. The flat parts match the period when the new cuticle hardens.

USE AFTER: END OF LESSON — End of lesson: exit ticket 'Brain Hormone, Ecdysone and Juvenile Hormone work together by...'

Q7. Which sequence correctly shows how hormones control a moult in an insect?

- A. Corpora allata, then Ecdysone, then Brain Hormone, then moult
- B. Brain Hormone, then the prothoracic gland releases Ecdysone, then moult; Juvenile Hormone decides the type of moult**
- C. Ecdysone, then the prothoracic gland, then Brain Hormone, then moult
- D. Juvenile Hormone, then Brain Hormone, then the prothoracic gland, then Ecdysone

Correct answer: B. Brain Hormone signals the prothoracic gland, which releases Ecdysone to trigger the moult. Juvenile Hormone from the corpora allata decides the kind of moult: larva to larva, larva to pupa or pupa to adult.